Healthy Skepticism Session 3

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Disclaimer

This presentation is for educational purposes only.

The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

Always seek the advice of a qualified physician or other licensed healthcare provider with any questions you may have regarding a medical condition, test result, or treatment. Never disregard professional medical advice or delay seeking it because of something discussed in this presentation.

The Five Faults

1. **Relative reporting** — Day 1
   What does a 36% improvement from taking a drug really mean?
2. **Probability reversal** — Day 2
   What does a positive COVID test really mean?
3. **Arbitrary categories** — Day 3
   How concerned should you be if your systolic BP goes from 119 to 120?
4. **The Bernoulli fallacy** — Day 3
   Why does the published medical advice keep changing?
5. **The causal leap** — Day 4
   Does bird watching prevent dementia?

Once you see them, you can't unsee them.

Today's Two Faults

3 **Arbitrary categories**

Blood pressure, body weight, and where the lines fall

4 **The Bernoulli fallacy**

The 0.05 line, and what a study can and cannot tell you

FAULT 3

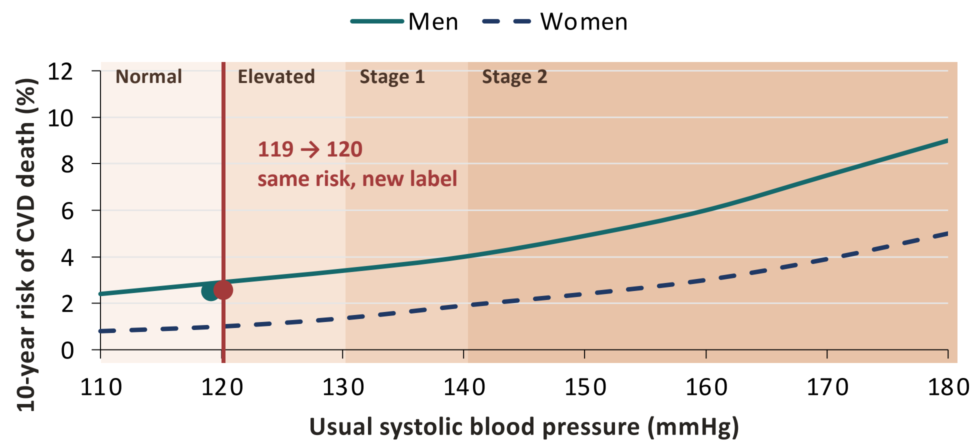
Arbitrary Categories

Read the curve, not the cutoff

Where Does “High” Begin?

- Systolic this year: 120. Last year: 119.
- Nothing about you changed. The label did.

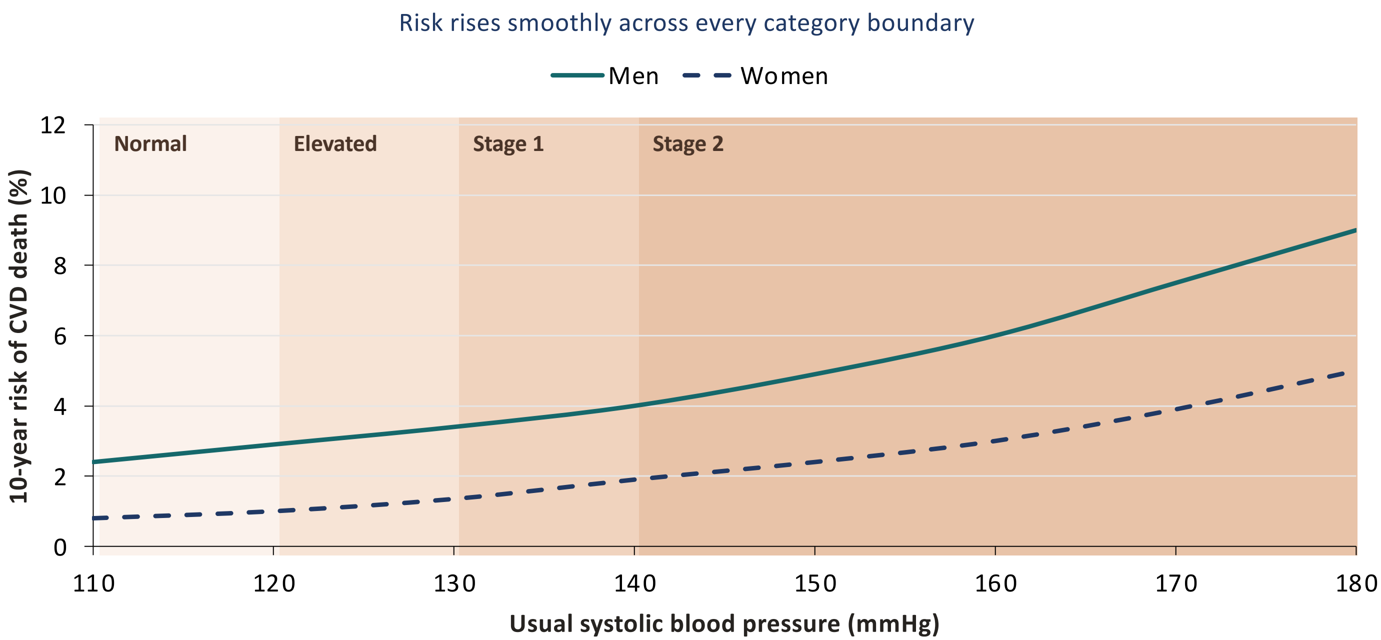
Risk climbs smoothly. The labels do not.



One mmHg moved you across a line. It did not move your risk.

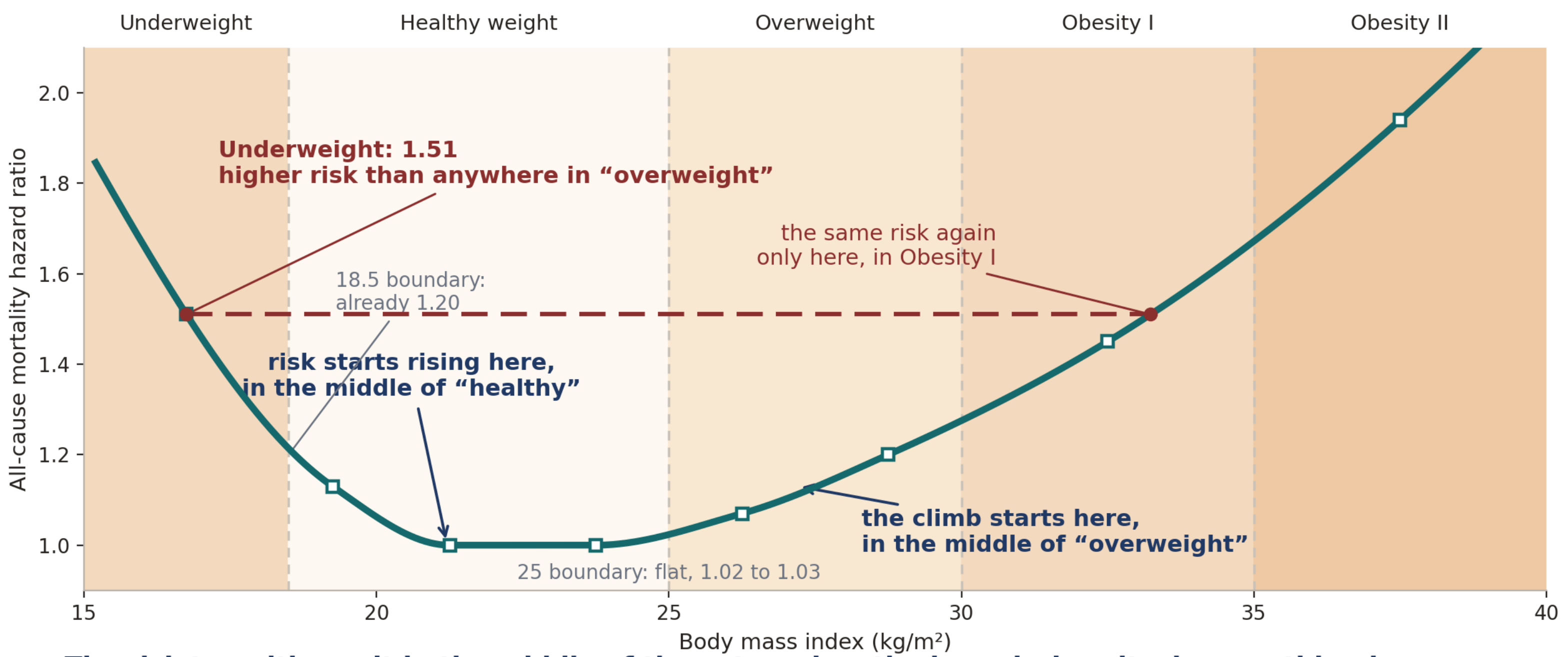
SCORE risk charts, low-risk regions: Conroy RM et al., Eur Heart J 2003;24:987-1003. Category boundaries: ACC/AHA 2017.

Systolic Categories Don't Mean Much



SCORE risk charts, low-risk regions: Conroy RM et al., Eur Heart J 2003;24:987-1003. Category boundaries: ACC/AHA 2017 hypertension guideline.

BMI Categories Are Worse



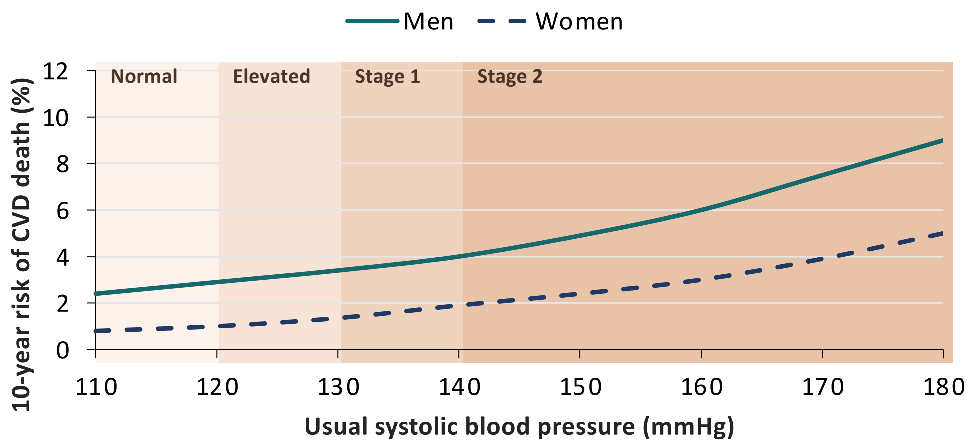
The risk transitions sit in the middle of the categories; the boundaries sit where nothing happens.

Global BMI Mortality Collaboration, Lancet 2016;388:776-786. All-cause mortality hazard ratios relative to BMI 22.5-25, never-smokers; band-midpoint values, shape-preserving interpolation.

So Why Have Categories at All?

- A box is a decision rule: triage in three seconds.
- The trouble: the box gets read as a fact about your body.
- A box cannot say how much risk you carry.

The risk curve does not know where the categories are



Ask where you sit on the curve, not which side of the line.

SCORE risk charts, low-risk regions: Conroy RM et al., Eur Heart J 2003;24:987-1003. Category boundaries: ACC/AHA 2017.

FAULT 4

The Bernoulli Fallacy

Why medical advice keeps changing

FAULT 4 · THE BERNOULLI FALLACY

Published Papers with Findings That Could be Repeated

Only 6 of 53 landmark cancer findings were confirmed on rerun.

Just 35 of 97 psychology findings stayed significant on rerun.

This is the Irreproducibility Crisis

Begley CG, Ellis LM. Nature 2012;483:531-533. Open Science Collaboration, Science 2015;349:aac4716. Clayton A, Bernoulli's Fallacy, 2021.

An Important Aside: P(A|B) is The Core Element of Inference

- It is called 'conditional probability'
- It means the probability of A if B is true
- Sometimes shortened to the probability of A, given B
- Depending on the circumstances, B may be observed, believed, or imagined for argument's sake
- **Examples:**
  - Courtrooms: P(Criminal guilt | evidence)
  - Gambling: P(Rolling 7 with 2 dice | Dice are fair)
  - Science: P(Tested hypothesis | data)
  - Causality: P(drowning | warm weather)

For the course, you only need to be aware of the concept

Sometimes We Want to Reverse Conditional Probability

Day 2 had 2 examples

- As we saw in Day 2, we can reverse P(A|B) to P(B|A) if we also have P(B)
- P(B) is sometimes called the 'prevalence' or 'prior' depending on the circumstances
- For example, in the mammogram example
  - P(called back | cancer) = 87%
  - P(cancer) = 0.6% (6 in 1000)
  - We computed using ratios P(cancer | called back) = 4.4%

This sort of calculation, reversing using the prior, is called 'Bayesian Analysis'

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The Fallacy Stems From Frequentist Reasoning

- They want to know if a hypothesis is true.
- Rather than compute P(hypothesis | data), they compute p = P(data | hypothesis is false)
- If p is low enough, they say the observations are 'statistically significant' and submit the result for publication.
- In medicine and the social sciences, the significance criterion is $p < .05$: if the hypothesis were false, the data would arise only 5% of the time.
  - In words: a false hypothesis would make this data very surprising, so treat it as true

This criterion means one would expect about 1 in 20 published experiments to be not reproducible.

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The Drawbacks

1. **The significance setting is arbitrary, and often far too high.** There is no formula or theorem that defines 'significance' .

2. **Significance is not informative.** Knowing P(hypothesis | data) enables risk-based decision support, like gambling; the p-value does not.

3. **It invites p-hacking.** Searching subsets of the data for those that meet the criterion by luck alone.

Together these feed the replication crisis

Five sigma, one-sided, corresponds to p = 0.000000287; ATLAS and CMS required it for the 2012 Higgs discovery. Ioannidis JPA. Why most published research findings are false. PLoS Med 2005;2:e124.

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Medical and Social Science Sets the p-value Far Too High for Significance

Medicine cannot wait for five-sigma data: patients are scarce, trials costly.

WHAT PHYSICS DEMANDS

$p < 0.000000287$

Five sigma: one chance in 3.5 million. The Higgs waited for that bar.

WHAT MEDICINE AND SOCIAL SCIENCE ACCEPT

$p < 0.05$

One chance in twenty. A no-effect test clears it 1 time in 20, by luck.

1 in 20 is the rate per test of nothing. The share of false findings is a reversal question: how many tested ideas were true? The prevalence, yet again.

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So Why Do They Adopt Frequentism?

- **Sometimes there is no prior:**
  - Industrial quality control
  - Census survey sampling
  - Particle physics (which sets the criterion at $p < 0.000000287$)
  - Not using a prior is considered more objective and honest by its proponents.
- **As described in the book**, a more rigorous standard disrupts the academic publication world.

Evidence of p-hacking

Only 6 of 53 (11%) landmark cancer findings were confirmed on rerun.

Just 35 of 97 (36%) psychology findings stayed significant on rerun.

These are much lower than chance (95%)

Begley CG, Ellis LM. Nature 2012;483:531-533. Open Science Collaboration, Science 2015;349:aac4716. Clayton A, Bernoulli's Fallacy, 2021.

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Evidence for ‘p-hacking’, aka ‘data torturing’

The observed failure rate is far above the honest floor.

1 in 20

the failure rate honest testing would produce on its own, with no torturing at all.

47 of 53

landmark cancer findings that failed to repeat. Psychology: about 2 in 3.

The failure rate is too high for luck alone.

Begley CG, Ellis LM. Nature 2012;483:531-533. Open Science Collaboration, Science 2015;349:aac4716.

FAULT 4 · THE BERNOULLI FALLACY

The Baby Aspirin Recommendation Change

Trials stopped early tend to stop on a lucky high. On the retests, p never crossed the line again.

1988

Physicians' Health Study: aspirin cut first heart attacks 44%, $p < 0.00001$. Stopped early for benefit. Decades of daily-aspirin advice followed.

2018

The retests, ARRIVE and ASPREE: no net benefit (ARRIVE $p = 0.60$), more major bleeding. USPSTF 2022: do not start after 60.

The advice changed because the finding would not repeat, and the harm outweighed the benefit

Physicians' Health Study final report, N Engl J Med 1989;321:129-135. McNeil JJ et al. (ASPREE), N Engl J Med 2018;379:1509-1518. Gaziano JM et al. (ARRIVE), Lancet 2018;392:1036-1046. USPSTF, JAMA 2022;327:1577-1584.

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Autism: From a p-Value to Policy

31	3x	$p < 0.05$
autism cases, parent-reported, not medical records	the reported odds, in boys only, not the full sample	so it was written up as a finding

1. **One study:** 31 parent-reported cases.
2. **One number:** p below 0.05.
3. **One citation:** the CDC's Autism and Vaccines page.
4. **One committee:** ACIP, twice, days before a vote.
5. **One policy:** universal newborn hepatitis B vaccination, ended.

Never replicated. In May 2026 the journal retracted the paper, months after the vote.

Gallagher CM, Goodman MS. J Toxicol Environ Health A 2010;73(24):1665-1677; cited on the CDC Autism and Vaccines page, since retracted. No replication; large studies find no association (Price CS et al., Pediatrics 2010;126:656-664).

What We've Learned

Two faults, one shape.

1

1 · Arbitrary categories

The trap: A line drawn across a continuous measurement is read as a fact about your body.

Ask: What is my risk with my measurement?

2

2 · The Bernoulli fallacy

The trap: p below 0.05 is read as proof the effect is real, and that has made the medical literature unreliable.

Ask: Have these studies been replicated?

Healthy Skepticism

Questions, and a conversation

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